

PREDICTION OF LUNG AND COLON CANCER BY USING TRANSFORMER GATED FUSION NETWORK

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Abstract—Cancer continues to be one of the main killers globally. A recent study found that the use of a hybrid model enables the detection of cancers in the lungs and colon through the processing of highly complicated histopathological images. The system integrates machine learning with deep learning through the use of ResNet18 to extract image features, which then interact with classifiers and Transformer networks for information fusion.

Hybrid techniques always perform better than single models. Integrating XGBoost with ResNet18 delivered a maximum accuracy of about 95.98, whereas deep learning models, such as CNN and Transformer for information fusion, attained 87.10. Maximum accuracy was delivered by integration of wavelets with CNN and Transformer, giving 97.95 accuracy. Therefore, feature types complement tumor classification. All these techniques have a promising future in diagnostics. Improving them is critical because this would decrease errors made by clinicians.

Index Terms—Cancer Classification, Histopathological Image Analysis, Hybrid Deep Learning, Convolutional Neural Networks, Transformer Networks, Gated Fusion Mechanism, Wavelet Transform, XGBoost

1. INTRODUCTION

The automated analysis of histopathology slides is revolutionizing oncology with rapid and accurate diagnosis of cancer. The two most prevalent causes of death globally include lung and colon cancer, with the time taken before initiating therapy being crucial for their prognosis. Early detection is a critical factor in enhancing patients' survival rates, while the existing manual method of examining tissue samples entails several challenges. The task requires substantial cognitive engagement and the need to recognize minuscule cellular features, making manual annotation highly cumbersome and subjective among pathologists.

Recently, artificial intelligence-based methods have been used in computational pathology to overcome these human limitations. CNNs are currently a gold standard, primarily due to their ability to capture multi-level spatial hierarchies of information similar to those of visual experts. However, CNN models usually ignore the global view of images and focus only on the local pixel level. This drawback has led to the development of ViT models that leverage self-attention mechanisms to comprehend long-term dependency patterns in data.

The combination of CNNs that recognize minute details at the local level with Transformers that establish the relationship between features at a broader and global level forms a potent combination for use in medical image processing. The hybrid architectures attempt to bridge the gap between cellular structures and larger tissue structures by establishing a more robust framework that can distinguish between benign and malignant tissues. In this paper, the development of an end-to-end classification model is explored using various machine learning and deep learning frameworks.

A. Motivation

The primary motivation behind the research presented here lies in the enormous global burden of cancer cases and death rates. Since the diagnosis of lung and colon cancers depends heavily on the timing of the discovery, fast and reliable methods of diagnosis are crucial for saving patients' lives. Traditionally, cancer diagnosis has been performed manually via microscopic examination, and despite being the most efficient way at some point in time, it is currently deemed insufficient due to being too slow and highly dependent on human efforts. Hence, there is an obvious necessity of developing new automated medical image analysis tools capable of processing large volumes of data quickly.

In terms of the development of computational intelligence, one can speak of the importance of creating hybrid approaches to artificial intelligence that allow for integrating feature extraction and broader reasoning. The idea of using different algorithms of machine learning is extremely helpful in solving the problem of algorithmic narrow-mindedness. Overall, the research presented here was prompted by the desire to create an approach to diagnosing cancer that would provide high levels of accuracy and precision, helping pathologists to perform their tasks easily.

B. Contribution

The main objective of the current investigation is to develop an enhanced hybrid model by integrating convolutional networks, Transformer layers, and Wavelet transformation to optimize histopathology image processing. In order to successfully integrate these diverse data sources, we propose a gated fusion method to effectively balance various feature vectors into one composite vector. The results of our empirical



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